

IMAT MOCK EXAM 10: Final Simulation

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 44/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Reading Skills, General Knowledge & Logical Reasoning (Q1-9)

1. "Stop taking this medicine and tell your doctor immediately if you notice any of the following rare side effects: swelling of the face, lips or throat, difficulty breathing, or a severe skin rash with blistering. Common side effects include nausea and headache. Uncommon side effects include dizziness and sleep disturbance. The frequency of certain other effects cannot be determined based on available data."

From the informational leaflet for a prescription antihistamine

Which of the following statements is the correct interpretation of the information contained in this excerpt?

- A) Sleep disturbance is a reason to stop the medicine immediately.
- B) Nausea is a rarer side effect of this medicine than a severe skin rash.
- C) A patient who develops a blistering skin rash while taking this medicine should stop taking it and contact a doctor immediately.
- D) Difficulty breathing is a common side effect of this medicine.
- E) Every patient who takes this medicine experiences at least one side effect.

Answer: C

The leaflet groups a severe skin rash with blistering, swelling of face, lips or throat, and difficulty breathing among the rare effects for which the instruction is to stop the medicine and contact a doctor immediately, so a patient who develops a blistering rash should do exactly that. Nausea is listed among the common effects, so it is more frequent than the rash, not rarer. Difficulty breathing belongs to the rare, urgent group, not to the common ones. The leaflet never suggests that every patient experiences side effects, and immediate stopping is reserved for the rare effects, while sleep disturbance is only uncommon.

2. "Street trees and small parks measurably cool dense neighbourhoods in summer. In a survey of ten Mediterranean cities, streets whose canopy covered more than 30% of the view from the pavement were on average 2.1 °C cooler on summer afternoons than otherwise similar streets with sparse canopy. The difference was largest in districts with low buildings and narrow streets, where shaded surfaces trap less heat. Cooling depended on species and on irrigation: unwatered young trees grew too slowly in dry summers to shade anything. City planners quoted in the report treat canopy as one instrument among several, alongside reflective roofs and lighter pavements."

From a newspaper report on urban climate research

Indicate which of the following statements is not correct:

- A) The report presents canopy as one of several instruments available to city planners.
- B) How well trees cool a street can depend on the species and on whether the trees are watered.
- C) Streets with more than 30% canopy cover were, on average, cooler than similar streets with sparse canopy in the survey.
- D) Tree planting has been shown by the report to be a complete solution to summer heat in Mediterranean cities.
- E) In districts with low buildings and narrow streets, the cooling difference associated with canopy was largest.

Answer: D

The report presents canopy as one instrument among several, alongside reflective roofs and lighter pavements, and never claims that planting trees alone solves summer heat, so the statement calling tree planting a complete solution goes beyond and against the passage and is the one that is not correct. The other statements faithfully restate the survey finding that streets with more than 30 percent canopy were on average 2.1 °C cooler, the dependence of cooling on species and irrigation, the largest effect in districts with low buildings and narrow streets, and the planners' multi-instrument view.

3. "In this country the ruler causes money to be made from the inner bark of mulberry trees, pressed into sheets like paper and cut into round pieces of different values. With this money everything can be bought throughout the whole empire. Refusing it is punishable, and every year the ruler causes all the mulberry bark that can be gathered to be bought up. It costs him almost nothing, yet with it he buys the goods of merchants from every land, who must accept it in exchange."

Adapted from The Travels of Marco Polo (13th century)

What does the passage describe?

- A) Gold coins stamped with the ruler's seal
- B) Barter between merchants and the ruler
- C) Letters of credit drawn on banks
- D) Seashells used as currency
- E) Paper money issued by a ruler

Answer: E

The passage describes money made from mulberry bark, pressed into sheets and cut into round pieces of different values, accepted throughout the empire on pain of punishment and bought up cheaply by the ruler; this is state-issued paper money. Gold coins are metal rather than bark sheets, letters of credit presuppose banks that do not appear in the text, barter would contradict the merchants' obligation to accept the money in exchange, and seashells are never mentioned.

4. "Bacteria in the large intestine ferment fibres that human digestive enzymes cannot break down. The fermentation releases short-chain fatty acids, which are absorbed and used by the cells of the intestinal wall as an energy source. Some of these acids also enter the bloodstream and influence the activity of immune cells. The composition of the bacterial population differs between people and shifts with diet over weeks."

From a popular science book on the human body

Only one of the following statements is coherent with the excerpt:

- A) Gut bacteria can convert dietary fibre into substances that the body absorbs and uses.
- B) Immune cell activity is independent of the products of gut bacteria.
- C) Short-chain fatty acids act only inside the intestinal lumen and never enter the blood.
- D) Human digestive enzymes release short-chain fatty acids from fibre in the small intestine.
- E) The bacterial population of the intestine is fixed at birth and does not change with diet.

Answer: A

The excerpt says that intestinal bacteria ferment fibre into short-chain fatty acids, which are absorbed, used as fuel by the cells of the intestinal wall and can enter the bloodstream to influence immune cells, so the coherent statement is that gut bacteria can convert dietary fibre into substances the body absorbs and uses. The claim that human enzymes release the acids in the small intestine contradicts the opening sentence, the population is said to shift with diet rather than be fixed at birth, the acids do enter the blood, and immune cell activity is influenced by them rather than independent of them.

5. Consider this statement: Every chemist working in this laboratory speaks German.

Which of the following statements is deducible from the given statement?

- A) Some chemists working in the laboratory do not speak German.
- B) Someone working in the laboratory who does not speak German is not one of its chemists.
- C) Nobody in the laboratory speaks German.
- D) Everyone in the laboratory who speaks German is a chemist.
- E) German is spoken in the laboratory only by chemists.

Answer: B

The given statement is 'every chemist working in this laboratory speaks German'. Its contrapositive must also be true: anyone working in the laboratory who does not speak German cannot be one of its chemists. The claim that everyone who speaks German is a chemist, and its restatement that German is spoken in the laboratory only by chemists, is the converse and does not follow; the claim that nobody speaks German contradicts the premise, and so does the claim that some chemists in the laboratory do not speak German.

6. From the given statements:

If the vaccine shipment arrives on Monday, then the clinic starts the vaccination campaign on Tuesday.

The campaign did not start on Tuesday.

Which conclusion can be deduced?

- A)** The campaign started on Monday.
- B)** The campaign will start on Wednesday.
- C)** The vaccine shipment was lost.
- D)** The vaccine shipment did not arrive on Monday.
- E)** The vaccine shipment arrived on Monday.

Answer: D

The premises have the form 'if the shipment arrives on Monday, then the campaign starts on Tuesday' together with 'the campaign did not start on Tuesday'. By modus tollens, the negation of the first part follows: the shipment did not arrive on Monday. That the shipment was lost goes beyond the premises, since a delay would fit equally well, and nothing is said about a Wednesday start or about the campaign starting on Monday.

7. Five runners, Ada, Beppe, Chiara, Dario and Elena, finished a race with no tied places. Dario finished before Chiara, Chiara finished before Ada, Ada finished before Beppe, and Beppe finished before Elena. Which of the following statements can be deduced?

- A)** Chiara finished last.
- B)** Ada finished before Dario.
- C)** Elena finished after Chiara.
- D)** Dario finished after Ada.
- E)** Beppe finished before Ada.

Answer: C

Chaining the constraints gives one unique order of finish: Dario, then Chiara, then Ada, then Beppe, then Elena. From this order, Elena finished after Chiara is necessarily true. The claims that Ada finished before Dario, that Beppe finished before Ada and that Dario finished after Ada each reverse a link of the chain, and Chiara finished last is false because Elena did.

8. Three friends, Nadia, Omar and Paolo, each drove home in a different car, one red, one blue and one white. Nadia drove neither the red nor the blue car, and Omar did not drive the blue car. Which of the following statements is true?

- A)** Omar drove the red car.
- B)** Nadia drove the blue car.
- C)** Paolo drove the red car.
- D)** Omar drove the white car.
- E)** Paolo drove the white car.

Answer: A

Nadia drove neither the red nor the blue car, so she took the white car. Omar did not drive the blue car and white is already taken, so Omar drove the red car, which makes that statement true, and Paolo is left with the blue car. Every step is forced, so the assignment is unique and each other statement (Nadia in blue, Paolo in white, Omar in white, Paolo in red) contradicts it.

9. A watch gains 3 minutes every hour. It is set to the correct time at 8:00 am. When the true time is 10:00 am, the watch shows:

- A) 9:57 am B) 10:00 am C) 10:03 am D) 10:06 am E) 10:09 am

Answer: D

Between a true time of 8:00 am and a true time of 10:00 am exactly 2 hours pass, and a watch gaining 3 minutes per hour gains $2 \times 3 = 6$ minutes over those 2 hours. The watch therefore shows 10:06 am. The reading 10:03 am counts only one hour of gain, 9:57 am treats the watch as losing time, and 10:09 am triples the gain.

Section B - Biology (Q10-32)

10. The immediate benefit that lactic fermentation provides to a muscle cell working without enough oxygen is that it:

- A) produces a large amount of ATP by oxidising lactate completely.
- B) transports pyruvate into the mitochondrial matrix.
- C) converts lactate into glycogen for storage.
- D) releases the energy of reduced FAD for ATP synthesis.
- E) restores NAD^+ so that glycolysis can continue producing ATP.

Answer: E

When oxygen is scarce, glycolysis converts NAD^+ to NADH, which would soon bring the pathway to a halt. Lactic fermentation reduces pyruvate to lactate and in doing so reoxidises NADH back to NAD^+ , letting glycolysis continue producing its small ATP yield; that regeneration is the immediate benefit to the working muscle cell. Complete oxidation of lactate for ATP, conversion of lactate into glycogen, energy from reduced FAD and transport of pyruvate into the mitochondrial matrix all belong to aerobic respiration or to later liver metabolism, not to fermentation inside the muscle fibre.

11. Which of the following statements about glycolysis is correct?

- A) It requires molecular oxygen in its first step.
- B) It takes place in the cytosol and produces a net gain of 2 ATP per glucose molecule.
- C) It reduces FAD to FADH_2 during the oxidation of glucose 6-phosphate.
- D) It takes place inside the mitochondrial matrix and produces a net gain of 4 ATP per glucose molecule.
- E) It converts glucose directly into acetyl-CoA without forming pyruvate.

Answer: B

Glycolysis splits glucose into two pyruvate molecules in the cytosol, producing 4 ATP while consuming 2 earlier in the pathway, for a net gain of 2 ATP per glucose, plus 2 NADH. No step requires molecular oxygen, FAD is not involved (FADH_2 appears later, in the Krebs cycle), and the product is pyruvate, not acetyl-CoA. The version placing glycolysis in the mitochondrial matrix with a net gain of 4 ATP is wrong on both counts.

12. In a eukaryotic cell supplied with oxygen, the largest share of the ATP produced from one glucose molecule comes from:

- A)** glycolysis itself.
- B)** lactic fermentation.
- C)** oxidative phosphorylation.
- D)** the conversion of pyruvate to acetyl-CoA.
- E)** the Krebs cycle operating directly.

Answer: C

Of the roughly 30 ATP obtained from one glucose molecule, glycolysis contributes a net 2 and the Krebs cycle 2 more by direct (substrate-level) phosphorylation, the link reaction contributes none, and fermentation adds nothing beyond glycolysis. The remaining share, about 26 ATP, arises when the NADH and FADH₂ generated by the earlier stages are oxidised through the electron transport chain, that is, by oxidative phosphorylation, which is therefore the largest contributor.

13. ATP synthase of the inner mitochondrial membrane produces ATP using the energy released by:

- A)** protons flowing down their gradient through the enzyme.
- B)** diffusion of glucose across the membrane.
- C)** direct transfer of a phosphate group from NADH to ADP.
- D)** hydrolysis of creatine phosphate.
- E)** electrons jumping directly onto oxygen molecules.

Answer: A

The electron transport chain pumps protons into the intermembrane space, storing energy in a gradient; protons then flow back down this gradient through ATP synthase, and the enzyme couples that flow to the phosphorylation of ADP. Oxygen only accepts electrons at the end of the chain and plays no part in ATP synthase, no phosphate group is transferred directly from NADH, creatine phosphate hydrolysis belongs to the muscle phosphagen system, and glucose diffusion across the membrane is irrelevant here.

14. In the mitochondrial respiratory chain, which carrier transfers electrons from complex I to complex III?

- A)** ATP synthase
- B)** NADH
- C)** Molecular oxygen
- D)** Cytochrome c
- E)** Ubiquinone (coenzyme Q)

Answer: E

Ubiquinone (coenzyme Q) is the small lipid-soluble mobile carrier that accepts electrons from complex I (and also from complex II) and delivers them to complex III. Cytochrome c performs the analogous shuttle but between complexes III and IV, NADH donates its electrons to complex I rather than moving between complexes, molecular oxygen only receives electrons at complex IV, and ATP synthase is not an electron carrier at all.

15. In a tissue that is metabolising actively and releasing much CO₂, haemoglobin releases oxygen more readily because:

- A)** the rising pH shifts haemoglobin into a high-affinity state.
- B)** H⁺ and CO₂ bind to haemoglobin and lower its affinity for oxygen.
- C)** oxygen bound to one heme group prevents oxygen from binding to the others.
- D)** CO₂ competes with oxygen for the iron atom of the heme group.
- E)** the red blood cells take up bicarbonate and expel all their oxygen.

Answer: B

In actively metabolising tissue, CO₂ forms carbonic acid and releases H⁺; both CO₂ and H⁺ bind to haemoglobin at sites away from the heme iron, stabilising its low-affinity state so that oxygen is unloaded more easily (the Bohr effect). A rising pH would have the opposite effect, since acidity lowers affinity; CO₂ does not compete with oxygen for the iron (carbon monoxide does); cooperative binding means oxygen binding promotes, rather than prevents, further binding; and red blood cells do not expel all their oxygen.

16. In skeletal muscle, the calcium ions released during contraction bind to troponin. The effect of this binding is to:

- A)** hydrolyse the ATP attached to the myosin heads.
- B)** provide the energy for the power stroke.
- C)** shorten the myosin filaments themselves.
- D)** move tropomyosin away from the myosin-binding sites on actin.
- E)** detach the myosin heads from actin permanently.

Answer: D

Calcium binding to troponin changes its shape and shifts the attached tropomyosin away from the myosin-binding sites on actin, exposing them so that cross-bridge formation can begin. The energy for the power stroke comes from ATP already hydrolysed on the myosin head, not from calcium; calcium binding neither detaches myosin heads (ATP binding does that) nor hydrolyses their ATP; and the filaments slide past each other without changing length, so the myosin filaments themselves do not shorten.

17. Cardiac muscle cells are joined end to end by intercalated discs containing gap junctions. The main function of these junctions is to:

- A)** allow rapid electrical coupling between adjacent cells.
- B)** store oxygen within the muscle cells.
- C)** increase the surface area available for gas exchange.
- D)** synthesise the collagen fibres of the heart skeleton.
- E)** anchor the cells firmly to the heart's tendons.

Answer: A

Gap junctions in the intercalated discs are protein channels that let ions pass directly from one cardiac cell to the next, so the electrical impulse begun at the pacemaker spreads rapidly and the muscle of a chamber contracts as a coordinated unit. Anchoring is the job of the desmosomes in the same discs, collagen synthesis belongs to connective tissue, these junctions have no gas-exchange role, and oxygen is stored inside the cells by myoglobin.

18. During contraction of a sarcomere, which region remains unchanged in length?

- A)** The I band
- B)** The I band together with the H zone
- C)** The A band
- D)** The H zone
- E)** None: every band shortens

Answer: C

The A band corresponds to the full length of the thick myosin filaments, which do not change length during contraction, so the A band stays constant. As the actin filaments slide inward, the I band (actin only) and the H zone (myosin with no overlapping actin) both narrow, and the sarcomere as a whole shortens, which is why the claim that every band shortens is false.

19. The effect of a competitive inhibitor on an enzyme-catalysed reaction can be overcome effectively by:

- A)** raising the pH to the enzyme's optimum value.
- B)** cooling the reaction mixture.
- C)** increasing the concentration of a cofactor of the enzyme.
- D)** adding a non-competitive inhibitor.
- E)** increasing the concentration of the substrate.

Answer: E

A competitive inhibitor binds reversibly in the active site and competes with substrate molecules for it, so raising the substrate concentration makes it more likely that substrate rather than inhibitor occupies the site, restoring the maximum rate. Changing the cofactor concentration, cooling the mixture or setting the pH to the optimum does not displace a molecule sitting in the active site, and adding a second inhibitor could only slow the reaction further.

20. A small organic non-protein molecule that must be bound to an enzyme for the enzyme to show its activity is called:

- A)** a holoenzyme.
- B)** a coenzyme.
- C)** a substrate analogue.
- D)** a zymogen.
- E)** an apoenzyme.

Answer: B

A small organic molecule that must be bound to an enzyme for activity is a coenzyme, examples being NAD⁺ and FAD. The protein alone without it is the apoenzyme, while the holoenzyme is the active complex of protein plus cofactor (the enzyme as a whole, not the small molecule); a zymogen is an inactive precursor of an enzyme, and a substrate analogue is a look-alike molecule that blocks the active site instead of enabling catalysis.

21. Which digestive enzyme acts in the stomach and has an optimum pH of about 2?

- A)** Pepsin
- B)** Pancreatic lipase
- C)** Trypsin
- D)** Intestinal maltase
- E)** Pancreatic amylase

Answer: A

Pepsin is secreted into the stomach, where hydrochloric acid activates it and gives it an optimum pH of about 2 for hydrolysing proteins into peptides. Pancreatic amylase, trypsin and pancreatic lipase all act at near-neutral to mildly alkaline pH in the small intestine, and intestinal maltase works at the neutral pH of intestinal juice, so none of them fits the stomach environment.

22. In the liver, the toxic ammonia produced by the breakdown of amino acids is converted mainly into:

- A)** uric acid.
- B)** creatinine.
- C)** glucose.
- D)** carbon dioxide.
- E)** urea.

Answer: E

In the urea cycle, run in liver cells, the ammonia released by amino acid breakdown is combined with carbon dioxide and an amino group to form urea, which the blood carries to the kidneys for excretion. Uric acid is the nitrogen waste of birds and reptiles, creatinine comes from phosphocreatine breakdown in muscle, and carbon dioxide and glucose are not products of ammonia detoxification.

23. If the secretion of antidiuretic hormone (ADH) stops, the most direct consequence for the kidneys is that the person produces:

- A)** urine containing glucose.
- B)** a small volume of highly concentrated urine.
- C)** urine with an abnormally high sodium concentration and no other change.
- D)** a large volume of dilute urine.
- E)** no urine at all.

Answer: D

ADH raises the water permeability of the collecting duct by promoting the insertion of aquaporin channels; if its secretion stops, water is no longer reabsorbed there, so a large volume of dilute urine is produced. A small volume of concentrated urine is the effect of ADH acting, not of its absence; glucose in the urine points to high blood glucose exceeding the renal threshold; and urine formation does not cease, only its volume and concentration change.

24. The cells that resorb bone matrix by secreting acid and digestive enzymes are:

- A)** osteoblasts. **B)** osteoclasts. **C)** chondroblasts. **D)** osteocytes. **E)** fibroblasts.

Answer: B

Osteoclasts attach to a bone surface, seal off a compartment and pump out acid and lysosomal enzymes, dissolving the mineral and digesting the organic matrix; this resorption releases calcium into the blood. Osteoblasts do the opposite by depositing new matrix, osteocytes are mature cells embedded in lacunae that maintain the tissue, chondroblasts form cartilage, and fibroblasts produce the fibres of connective tissue.

25. Which hormone raises the concentration of glucose in the blood by stimulating glycogen breakdown in the liver?

- A)** Melatonin **B)** Insulin **C)** Glucagon **D)** Aldosterone **E)** Thyroxine

Answer: C

Glucagon, released by the alpha cells of the pancreatic islets when blood glucose is low, activates glycogen phosphorylase in the liver, so stored glycogen is broken down and glucose is released into the blood. Insulin has the opposite action, promoting glycogen synthesis and glucose uptake; aldosterone regulates sodium and potassium balance, melatonin governs circadian rhythms, and thyroxine raises the basal metabolic rate rather than serving as the glucose-raising hormone described.

26. The bond that joins two glucose units within glycogen is:

- A)** a peptide bond.
B) a phosphodiester bond.
C) a glycosidic bond.
D) a hydrogen bond.
E) an ester bond.

Answer: C

In glycogen, as in starch and cellulose, glucose units are joined by glycosidic bonds, formed by condensation between hydroxyl groups of adjacent sugars. Peptide bonds join amino acids in proteins, ester bonds link glycerol to fatty acids in fats, phosphodiester bonds join nucleotides in nucleic acids, and hydrogen bonds only stabilise structures weakly without covalently linking the sugar units.

27. A cell is treated with ouabain, a poison that blocks the sodium-potassium pump. The most direct consequence is that the cell gradually:

- A)** loses the Na^+/K^+ concentration gradient across its membrane.
B) reverses the direction of the Na^+/K^+ gradient across its membrane.
C) increases its rate of ATP production.
D) starts pumping potassium out and sodium in.
E) becomes freely permeable to all ions.

Answer: A

The sodium-potassium pump continuously exports 3 Na^+ and imports 2 K^+ per ATP hydrolysed, maintaining the concentration gradients against constant leakage through the membrane. Blocking it stops this maintenance, so the gradients gradually decay and are lost. The pump neither runs backwards nor reverses the gradients, the membrane does not become freely permeable to all ions, and with the pump idle the cell consumes less ATP, not more.

28. A digestive enzyme secreted by a pancreatic cell reaches the plasma membrane by passing through which sequence of structures?

- A)** Golgi apparatus, rough ER, lysosome, plasma membrane.
B) Smooth ER, Golgi apparatus, nucleus, plasma membrane.
C) Free ribosome, then nucleus, then plasma membrane.
D) Rough ER, Golgi apparatus, secretory vesicle, plasma membrane.
E) Rough ER, lysosome, Golgi apparatus, plasma membrane.

Answer: D

A digestive enzyme is a secreted protein: it is synthesised by ribosomes attached to the rough endoplasmic reticulum, enters the ER lumen, travels in transport vesicles to the Golgi apparatus for modification and sorting, and leaves in secretory vesicles that fuse with the plasma membrane. Free ribosomes make proteins that stay in the cytosol, and the secretory route never passes through the nucleus or the lysosomes.

29. Lysosomal enzymes are synthesised on the rough ER and are then sorted, tagged and packaged for delivery to the lysosomes by:

- A)** the nucleolus.
- B)** the Golgi apparatus.
- C)** the peroxisomes.
- D)** the smooth endoplasmic reticulum.
- E)** the mitochondria.

Answer: B

After synthesis on the rough ER, lysosomal enzymes pass to the Golgi apparatus, where they are tagged (typically with phosphate groups on mannose sugars) and packed into vesicles destined for the lysosomes. The smooth ER handles lipid synthesis and detoxification, the nucleolus assembles ribosomal subunits, and peroxisomes and mitochondria are separate organelles with their own enzymes, none of which sorts lysosomal cargo.

30. A triglyceride molecule consists of:

- A)** three amino acids joined to glycerol.
- B)** one glucose unit joined to three fatty acids.
- C)** glycerol and four fatty acids.
- D)** glycerol, two fatty acids and one phosphate group.
- E)** glycerol and three fatty acids.

Answer: E

A triglyceride is one glycerol molecule with its three hydroxyl groups esterified to three fatty acid chains, the form in which fat is stored in adipose tissue. Replacing one fatty acid with a phosphate group gives a phospholipid instead; amino acids and glucose are not components of triglycerides, and four fatty acids would not fit the three hydroxyls of glycerol.

31. The ability to taste a certain bitter compound is controlled by a dominant allele T. A man who is a heterozygous taster (Tt) and a woman who cannot taste the compound (tt) have a child. The probability that the child cannot taste the compound is:

- A)** 0 **B)** 1/4 **C)** 1/2 **D)** 1 **E)** 3/4

Answer: C

The father is Tt and the mother is tt. The father's gametes carry T or t with equal chance, while every maternal gamete carries t, so half the children are Tt (tasters) and half are tt (non-tasters). The probability of a child who cannot taste the compound is therefore 1/2. The fractions 1/4 and 3/4 arise from a Tt × Tt cross, which is not the cross given here.

32. During quiet inhalation, which of the following occurs?

- A)** The diaphragm contracts and the thoracic volume increases.
- B)** The diaphragm relaxes and rises back into the thorax.
- C)** The internal intercostal muscles contract strongly.
- D)** The pressure inside the alveoli rises above atmospheric pressure.
- E)** The rib cage is pulled inward.

Answer: A

During quiet inhalation the diaphragm contracts and flattens and the external intercostal muscles raise the ribs, enlarging the thoracic cavity; alveolar pressure then falls slightly below atmospheric pressure and air flows in. A diaphragm that relaxes and rises back into the thorax, strongly contracting internal intercostals, a rib cage pulled inward, and alveolar pressure above atmospheric all describe exhalation.

Section C - Chemistry (Q33-47)

33. A 0.001 mol/dm^3 solution of a monoprotic weak acid has a pH of 4.0. The acid dissociation constant K_a of this acid is approximately:

- A)** 1.0×10^{-3} **B)** 1.0×10^{-4} **C)** 1.0×10^{-5} **D)** 1.0×10^{-6} **E)** 1.0×10^{-7}

Answer: C

A pH of 4.0 means $[\text{H}^+] = 1.0 \times 10^{-4} \text{ mol/dm}^3$, and dissociation of the monoprotic acid HA gives equal concentrations of H^+ and A^- . Using $K_a = [\text{H}^+][\text{A}^-]/[\text{HA}]$ with $[\text{HA}] \approx \text{C}$: $K_a \approx (10^{-4} \times 10^{-4})/10^{-3} = 10^{-8}/10^{-3} = 1.0 \times 10^{-5}$ (the exact denominator $10^{-3} - 10^{-4} = 9 \times 10^{-4}$ gives 1.1×10^{-5} , the same approximation). Equating K_a with $[\text{H}^+]$ itself would wrongly give 10^{-4} , and multiplying $[\text{H}^+]$ by the concentration instead of squaring and dividing would give 10^{-7} .

34. An aqueous solution of ammonium chloride, NH_4Cl , the salt of a strong acid and a weak base, is:

- A)** impossible to classify without knowing the concentration.
B) acidic, because NH_4^+ donates a proton to water.
C) basic, because Cl^- is a strong base.
D) neutral, because it is a salt and salts do not affect pH.
E) basic, because NH_4^+ releases hydroxide ions.

Answer: B

NH_4Cl is the salt of the weak base NH_3 and the strong acid HCl . In water the Cl^- ion, as the conjugate base of a strong acid, is a negligible spectator, while NH_4^+ , the conjugate acid of a weak base, donates a proton to water producing H_3O^+ , so the solution is acidic. Salts are not always neutral, Cl^- is not a strong base, NH_4^+ releases protons rather than hydroxide ions, and the concentration would change the degree of acidity but not its direction.

35. A solid substance does not conduct electricity at all, but the same substance in the molten state conducts well. The substance is most likely:

- A)** glucose.
B) sulfur.
C) copper.
D) sodium chloride.
E) silicon dioxide.

Answer: D

No conduction as a solid but good conduction when molten is the signature of an ionic solid: in the crystal the ions are fixed in the lattice, while melting frees them to move and carry charge. This fits sodium chloride. Copper conducts even as a solid through its delocalised electrons, and sulfur, silicon dioxide and glucose, being molecular substances, conduct in neither state.

36. Which of the following bonds is the most polar? (Pauling electronegativities: H = 2.20, C = 2.55, N = 3.04, O = 3.44, Cl = 3.16)

- A)** C-N **B)** C-Cl **C)** C-H **D)** C-C **E)** C-O

Answer: E

Bond polarity increases with the electronegativity difference between the two atoms. The differences from carbon (2.55) are: with oxygen, $3.44 - 2.55 = 0.89$; with chlorine, $3.16 - 2.55 = 0.61$; with nitrogen, $3.04 - 2.55 = 0.49$; with hydrogen, $2.55 - 2.20 = 0.35$; with another carbon, 0. The largest difference (0.89) belongs to the carbon-oxygen bond, making it the most polar; chlorine and nitrogen follow, and the carbon-carbon bond is non-polar.

37. What is the average oxidation number of sulfur in sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$?

- A) -2 B) 0 C) +2 D) +4 E) +6

Answer: C

The compound is neutral, so the oxidation numbers must sum to zero. Sodium contributes $2 \times (+1)$ and oxygen $3 \times (-2)$, so $2 + 2x - 6 = 0$ for the two sulfur atoms together, giving $2x = +4$ and an average of +2 per sulfur. The value +6 corresponds to sulfur in a sulfate ion and -2 to sulfide; in thiosulfate the two sulfur atoms actually have different individual values, and only their average is +2.

38. In the reaction $2\text{Mg} + \text{CO}_2 \rightarrow 2\text{MgO} + \text{C}$, which species is oxidised?

- A) Magnesium, because its oxidation number rises from 0 to +2.
B) CO_2 , because it loses oxygen atoms.
C) Carbon, because its oxidation number falls from +4 to 0.
D) Oxygen, because it changes partner.
E) Both magnesium and carbon are oxidised.

Answer: A

Oxidation is electron loss, seen as a rise in oxidation number. Magnesium goes from 0 in the element to +2 in MgO , so magnesium is oxidised, and its electrons reduce carbon from +4 in CO_2 to 0 in the deposited carbon. Describing carbon as oxidised gets it backwards (falling from +4 to 0 is reduction), oxygen merely changes partner while staying at -2, and both magnesium and carbon cannot be oxidised at once because one partner of the electron transfer must be reduced.

39. Which of the following compounds is a ketone?

- A) Propan-1-ol B) Methanal C) Propanal D) Propanoic acid E) Propanone

Answer: E

A ketone carries its carbonyl group, $\text{C}=\text{O}$, on a carbon inside the carbon chain. Propanone, CH_3COCH_3 , is the three-carbon ketone (acetone). Propanal and methanal have the carbonyl at the end of the chain and are aldehydes, propan-1-ol is an alcohol, and propanoic acid is a carboxylic acid.

40. Which of the following pairs contains only monosaccharides?

- A) glucose and sucrose
B) starch and cellulose
C) maltose and fructose
D) glucose and fructose
E) lactose and galactose

Answer: D

Monosaccharides are single sugar units: glucose, fructose and galactose qualify. Sucrose (glucose plus fructose), maltose (two glucose units) and lactose (galactose plus glucose) are disaccharides, and starch and cellulose are polysaccharides. Only the pair glucose and fructose consists of two monosaccharides.

41. What amount of oxygen, in moles, is required for the complete combustion of 0.5 mol of propane? ($\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$)

- A) 0.5 mol B) 1.5 mol C) 2.0 mol D) 2.5 mol E) 5.0 mol

Answer: D

The balanced equation requires 5 mol of O_2 per 1 mol of propane, so 0.5 mol of propane needs $0.5 \times 5 = 2.5$ mol of oxygen. The values 1.5 mol and 2.0 mol confuse the oxygen demand with the amounts of CO_2 (3×0.5) and H_2O (4×0.5) formed, 0.5 mol assumes a one-to-one ratio, and 5.0 mol forgets to scale from 1 mol down to 0.5 mol.

42. What mass of water is produced when 4.0 g of methane burns completely in excess oxygen? ($\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$; molar masses: $\text{CH}_4 = 16 \text{ g/mol}$, $\text{H}_2\text{O} = 18 \text{ g/mol}$)

- A)** 4.5 g **B)** 8.0 g **C)** 9.0 g **D)** 18.0 g **E)** 36.0 g

Answer: C

Step 1: amount of methane, $n = 4.0 \text{ g} \div 16 \text{ g/mol} = 0.25 \text{ mol}$

Step 2: the equation yields 2 mol of water per 1 mol of methane, so water formed = $0.25 \times 2 = 0.5 \text{ mol}$.

Step 3: mass = $0.5 \times 18 \text{ g/mol} = 9.0 \text{ g}$

The value 4.5 g forgets the factor of 2 in front of H_2O , while 18.0 g and 36.0 g lose track of how little methane is burning.

43. Going from left to right across a period of the periodic table, the first ionisation energy of the elements generally:

- A)** decreases, because the outer electron lies further from the nucleus.
B) increases, because nuclear charge rises while the outer electrons enter the same shell.
C) increases, because the atomic radius increases.
D) remains constant, because the shell number does not change.
E) decreases, because the atoms become heavier.

Answer: B

Across a period the nuclear charge rises while the added electrons enter the same principal shell, so the outermost electron is held more strongly and the first ionisation energy generally increases. That atoms become heavier does not by itself lower the ionisation energy, a constant shell number does not make the trend constant, and the atomic radius actually decreases across a period, so any explanation built on a growing radius is wrong.

44. The maximum number of electrons that the 3p sublevel can hold is:

- A)** 2 **B)** 3 **C)** 6 **D)** 8 **E)** 10

Answer: C

A p sublevel consists of three orbitals, and each orbital holds at most two electrons of opposite spin, giving $3 \times 2 = 6$ electrons. The value 2 is the capacity of a single orbital, 10 is the capacity of a d sublevel (five orbitals), and 8 is the capacity of a complete shell such as $n = 2$.

45. The industrial synthesis of ammonia is exothermic: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, $\Delta H < 0$. Which set of conditions gives the highest equilibrium yield of NH_3 ?

- A)** Low temperature and low pressure
B) Any temperature, provided a catalyst is used
C) High temperature and low pressure
D) High temperature and high pressure
E) Low temperature and high pressure

Answer: E

By Le Chatelier's principle, raising the pressure shifts the equilibrium toward the side with fewer moles of gas; here 4 mol of gas become 2, so high pressure favours NH_3 . Because the forward reaction is exothermic, lowering the temperature also favours the products (at the cost of rate). The highest yield therefore comes from low temperature and high pressure. High temperature favours the reverse (endothermic) direction, and a catalyst only speeds the approach to equilibrium without changing its position.

46. Adding a catalyst to a reversible reaction already at equilibrium:
- A) shifts the equilibrium position towards the products.
 - B) lowers the activation energies of forward and reverse reactions and does not shift the position of equilibrium.
 - C) raises the activation energy of the forward reaction only.
 - D) increases the concentrations of the reactants.
 - E) increases the enthalpy change of the reaction.

Answer: B

A catalyst provides an alternative reaction path of lower activation energy, and it lowers the barrier for the forward and reverse reactions equally, so both rates rise by the same factor and the equilibrium composition is unchanged; equilibrium is simply reached sooner. It does not shift the position toward products, it does not alter the enthalpy change (which depends only on the initial and final states), and it does not increase reactant concentrations.

47. The compound CuSO_4 is correctly named:
- A) copper(II) sulfate.
 - B) copper(II) sulfide.
 - C) copper(I) sulfate.
 - D) copper(I) sulfide.
 - E) copper(II) sulfite.

Answer: A

The sulfate ion SO_4^{2-} carries a 2– charge, so the single metal ion in CuSO_4 must be Cu^{2+} , and the Stock name is copper(II) sulfate. Copper(I) would need two copper ions per sulfate, giving the formula Cu_2SO_4 ; sulfite is SO_3^{2-} (one oxygen fewer than sulfate) and sulfide is the bare S^{2-} ion, so the copper sulfite and copper sulfide names refer to different compounds altogether.

Section D - Mathematics (Q48-54)

48. The set of real solutions of the inequality $x^2 + 2x + 5 \leq 0$ is:
- A) $x \leq -1$.
 - B) the set of all real numbers.
 - C) only $x = -1$.
 - D) the empty set.
 - E) $x \geq -1$.

Answer: D

Completing the square gives $x^2 + 2x + 5 = (x + 1)^2 + 4$

Since $(x + 1)^2 \geq 0$ for every real x , the expression is at least 4 and is therefore always positive; equivalently, the discriminant is $2^2 - 4 \times 1 \times 5 = -16$, negative, so the upward-opening parabola never touches the x -axis and no interval of solutions exists. The solution set is empty; in particular $x = -1$, the vertex position, gives the value 4 and does not satisfy the inequality.

49. Solve the inequality $5x - 3 < 2x + 9$.
- A) $x > 4$ B) $x < 2$ C) $x > 2$ D) $x < 12$ E) $x < 4$

Answer: E

Subtracting $2x$ from both sides gives $3x - 3 < 9$; adding 3 gives $3x < 12$; dividing by the positive number 3, which does not flip the inequality, gives $x < 4$. The answer $x < 12$ forgets the division by 3, and no sign reversal occurs anywhere because the inequality is never multiplied or divided by a negative number.

50. The equation $x^2 - 5x + k = 0$ has one root equal to 2. The value of k is:

- A) -6 B) 2 C) 4 D) 6 E) 10

Answer: D

A root satisfies the equation, so substituting $x = 2$ gives $2^2 - 5 \times 2 + k = 4 - 10 + k = 0$, hence $k = 6$. As a check, $x^2 - 5x + 6 = 0$ has roots 2 and 3, which sum to 5 and multiply to 6, consistent with the coefficients. The value 4 merely copies the square of the given root, and 10 copies the coefficient of x .

51. If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, how many elements does $A \cup B$ contain?

- A) 4 B) 5 C) 6 D) 7 E) 8

Answer: C

The union collects every element of both sets, counting shared elements once: $A \cup B = \{1, 2, 3, 4, 5, 6\}$. In symbols, $|A \cup B| = |A| + |B| - \text{shared} = 4 + 4 - 2 = 6$, since 3 and 4 belong to both. The answer 8 double counts the two shared elements, and 4 counts only one of the two sets.

52. The product $(2x - 3)(x + 5)$ is equal to:

- A) $2x^2 + 7x - 15$ B) $2x^2 - 7x - 15$ C) $x^2 + 7x - 15$ D) $2x^2 + 13x - 15$ E) $2x^2 + 7x + 15$

Answer: A

Multiplying term by term: $2x \times x = 2x^2$, $2x \times 5 = 10x$, $(-3) \times x = -3x$, and $(-3) \times 5 = -15$

Combining the middle terms, $10x - 3x = 7x$, gives $2x^2 + 7x - 15$

The version with $13x$ adds $10x$ and $3x$ instead of subtracting, and the forms with $-7x$, a positive 15 or a missing factor of 2 on the leading term come from sign and coefficient slips.

53. A rectangle's length is 3 cm greater than its width, and its perimeter is 26 cm. Its width is:

- A) 4 cm B) 5 cm C) 6 cm D) 8 cm E) 10 cm

Answer: B

Let the width be w ; the length is then $w + 3$.

The perimeter is $2(w + w + 3) = 26$, so $2w + 3 = 13$, $2w = 10$ and $w = 5$ cm, with length 8 cm

The value 8 cm reports the length rather than the width.

54. If $\sin \theta = 0.6$ and θ is an acute angle, then $\cos \theta$ is:

- A) 0.36 B) 0.4 C) 0.6 D) 0.8 E) 1.0

Answer: D

From the identity $\sin^2 \theta + \cos^2 \theta = 1$: $\cos^2 \theta = 1 - (0.6)^2 = 1 - 0.36 = 0.64$, and because θ is acute the cosine is positive, so $\cos \theta = \sqrt{0.64} = 0.8$. The value 0.36 is just $\sin^2 \theta$ itself, 0.4 subtracts 0.6 from 1 without squaring first, and 0.6 would simply repeat the sine.

Section E - Physics (Q55-60)

55. A car travels along a straight motorway at a constant 90 km/h, following a truck that travels at a constant 72 km/h and is 2.0 km ahead. After 5 minutes, which statement is correct?

- A) The car needs more than a further hour to catch the truck.
- B) The car has overtaken the truck.
- C) The car has drawn exactly level with the truck.
- D) The car will never catch the truck.
- E) The car has not yet caught the truck.

Answer: E

The car closes the gap at the relative speed $90 - 72 = 18$ km/h. In 5 minutes, which is $1/12$ of an hour, it gains $18 \times 1/12 = 1.5$ km, less than the initial 2.0 km gap, so it still trails by 0.5 km and has not yet caught the truck. Catching up completely would take $2.0 \div 18$ of an hour, about 6.7 minutes, so the car does catch the truck soon after but is not level at the 5 minute mark.

56. A circuit contains two resistors of $6\ \Omega$ each connected in parallel, and this pair is connected in series with a single $3\ \Omega$ resistor. The equivalent resistance of the whole network is:

- A) $2\ \Omega$ B) $3\ \Omega$ C) $6\ \Omega$ D) $9\ \Omega$ E) $15\ \Omega$

Answer: C

Two equal resistors in parallel have half the resistance of one: $6 \div 2 = 3\ \Omega$. This pair is in series with the $3\ \Omega$ resistor, and series resistances add: $3 + 3 = 6\ \Omega$. Treating all three as a series chain gives $15\ \Omega$, and ignoring one of the parallel pair gives $9\ \Omega$; both misread the network.

57. A wire of uniform material is replaced by a new wire that is twice as long and has half the cross-sectional area. Compared with the original wire, the resistance of the new wire is:

- A) unchanged.
- B) four times greater.
- C) half.
- D) four times smaller.
- E) double.

Answer: B

For a uniform wire, $R = \rho L/A$, so resistance is proportional to length and inversely proportional to cross-sectional area. Doubling the length doubles R , and halving the area doubles R again, giving $2 \times 2 = 4$ times the original resistance. The answers unchanged and double each account for only one of the two changes, and four times smaller reverses the direction of the effect.

58. At what height above the Earth's surface is the acceleration due to gravity equal to one quarter of its value at the surface? (The Earth's radius is R .)

- A) R B) $2R$ C) $R/4$ D) $R/2$ E) $4R$

Answer: A

At distance r from the Earth's centre, g is proportional to $1/r^2$. Requiring g to fall to one quarter of its surface value gives $(R/(R + h))^2 = 1/4$, so $R/(R + h) = 1/2$ and $R + h = 2R$, hence $h = R$: a height of one Earth radius above the surface. The answers $R/2$ and $2R$ come from forgetting to take the square root or from applying the factor 2 to the height instead of the distance from the centre.

59. A car of mass 1.5×10^3 kg accelerates along a level road at 2.0 m/s^2 against a constant frictional force of 500 N. The force the engine must exert is:

- A)** 500 N **B)** 2500 N **C)** 3000 N **D)** 3500 N **E)** 4000 N

Answer: D

The net force needed for the acceleration follows from Newton's second law: $F(\text{net}) = ma = 1.5 \times 10^3 \times 2.0 = 3000 \text{ N}$. The engine must in addition overcome the 500 N friction, so the driving force it must exert is $3000 + 500 = 3500 \text{ N}$. The value 3000 N omits friction, and 2500 N would correspond to subtracting friction from the net force, which would make no physical sense.

60. A swimming pool is 4 m deep. Taking the density of water as 1000 kg/m^3 and $g = 10 \text{ m/s}^2$, the gauge pressure of the water at the bottom of the pool is:

- A)** 400 Pa **B)** 2500 Pa **C)** 4000 Pa **D)** 40000 Pa **E)** 400000 Pa

Answer: D

Gauge pressure at depth h is $P = \rho gh = 1000 \text{ kg/m}^3 \times 10 \text{ m/s}^2 \times 4 \text{ m} = 40000 \text{ Pa}$ ($4.0 \times 10^4 \text{ Pa}$)

Because the question asks for gauge pressure, atmospheric pressure is excluded. The values 4000 Pa and 400000 Pa are each off by one power of ten, and 400 Pa by three.